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SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS: 5

Total Marks: 40

82%
48

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies 2.5	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30%	Well-planned, systematic implementation with optimal solution 3	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20%	Effective and advanced use of tools/technologies with proper configuration 2	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15%	Insightful analysis with accurate interpretation and validation of results 1.5	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10%	Clear, well-structured documentation with diagrams, code, and results 1	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

	Understanding of Concepts (25%)	Experimental Design (30%)	Use of Tools (30%)	Analysis of Results (15%)	Documentation (10%)
Q1	2.5	3	2.1	1.5	1
Q2	2	2.5	1	1	0.5
Q3	2.5	3	1	1.5	1
Q4	2	3	1	1	1
Q5	2.5	2.5	1	1.5	0.5

9
7
9
8
8

41

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1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

Question 1: LAN with Two PCs via a Switch — Ethernet Frame Analysis

Aim

To design a simple LAN consisting of two PCs connected through a switch in Cisco Packet Tracer, configure IP addressing, verify connectivity using the ping command, and capture and analyze the Ethernet frame fields exchanged between the hosts using Wireshark.

Procedure

1. Open Cisco Packet Tracer and place two end devices (PC-A and PC-B) and one 2960 switch onto the workspace.
2. Connect PC-A to switch port Fa0/1 and PC-B to switch port Fa0/2 using straight-through copper cables.
3. Assign IP address 192.168.1.1/24 to PC-A and 192.168.1.2/24 to PC-B through Desktop > IP Configuration.
4. Open the Command Prompt on PC-A and verify Layer-2/Layer-3 connectivity by executing ping 192.168.1.2.
5. Enable Wireshark (or Packet Tracer's Simulation mode) on the link between PC-A and the switch to capture the frames generated during the ping.
6. Open a captured frame and record the Source MAC, Destination MAC, EtherType, and FCS fields from the Ethernet II header.

Topology Diagram

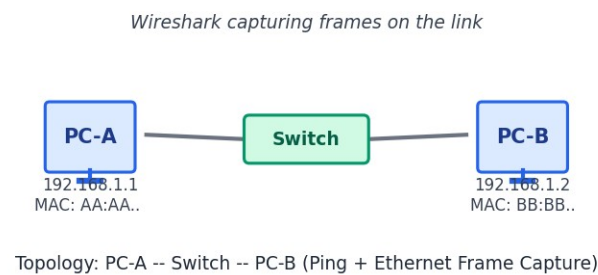


Fig. 1 — PC-A and PC-B interconnected through a Layer-2 switch

Wireshark Observation — Ethernet II Frame Fields

Field	Captured Value	Description
Source MAC Address	AA:AA:AA:AA:AA:01 (PC-A NIC)	Hardware address of the sending network adapter
Destination MAC Address	BB:BB:BB:BB:BB:02 (PC-B NIC)	Hardware address of the intended receiving adapter
EtherType	0x0800 (IPv4) / 0x0806 (ARP)	Identifies the encapsulated upper-layer protocol
Frame Check Sequence (FCS)	4-byte CRC, e.g. 0x3F2A91C4	Used by the receiver to detect transmission errors
Ping Result	4 packets sent, 4 received, 0% loss	Confirms successful end-to-end Layer-2 connectivity

Analysis / Explanation

Every device on an Ethernet LAN is identified by a unique 48-bit MAC address burned into its NIC. When PC-A pings PC-B, the Data Link layer encapsulates the IP packet inside an Ethernet II frame and stamps it with PC-A's MAC address as the source and PC-B's MAC address as the destination. Because PC-A does not initially know PC-B's MAC address, it first broadcasts an ARP request (EtherType 0x0806) to resolve the IP-to-MAC mapping; once resolved, subsequent frames carry EtherType 0x0800 for IPv4 traffic. The switch reads only the destination MAC address and forwards the frame out of the correct port using its MAC address table (learned by examining source addresses of incoming frames), rather than flooding it to every device once the mapping is known. The FCS field, computed using a Cyclic Redundancy Check (CRC-32), allows the receiving NIC to verify that the frame was not corrupted in transit; a mismatch causes the frame to be silently discarded. Together, MAC addressing and switch-based forwarding allow the Data Link layer to guarantee accurate, loop-free, and efficient frame delivery within a single local network segment.

2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

Question 2: Two LANs via a Router — IPv4 Header Analysis

Aim

To create a topology consisting of two LANs interconnected through a router, configure IPv4 addressing, subnet masks and default gateways, verify end-to-end communication, and analyze the IPv4 header fields captured in Wireshark.

Procedure

1. Build the topology: PC-A — Switch1 — Router (Fa0/0) and Router (Fa0/1) — Switch2 — PC-B in Cisco Packet Tracer.
2. Configure LAN 1 with network 192.168.1.0/24 and LAN 2 with network 192.168.2.0/24.
3. Assign PC-A the IP 192.168.1.10/24 with gateway 192.168.1.1, and PC-B the IP 192.168.2.10/24 with gateway 192.168.2.1.
4. Configure the router's two interfaces with 192.168.1.1/24 and 192.168.2.1/24 respectively, and issue 'no shutdown'.
5. From PC-A, ping PC-B (ping 192.168.2.10) to verify inter-network connectivity end to end.
6. Capture the ICMP packets in Wireshark and examine the IPv4 header — Source IP, Destination IP, TTL, Protocol Number, and Header Length.

Topology Diagram

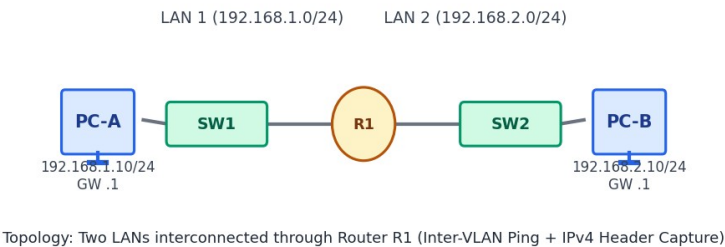


Fig. 2 — Two LANs interconnected through Router R1

Wireshark Observation — IPv4 Header Fields

Field	Captured Value	Description
Source IP Address	192.168.1.10	Address of the originating host (PC-A)
Destination IP Address	192.168.2.10	Address of the target host (PC-B)
Time-To-Live (TTL)	127 (decremented from 128 by router)	Hop limit; prevents infinite routing loops
Protocol Number	1 (ICMP)	Identifies the encapsulated Layer-4/ICMP protocol
Header Length (IHL)	20 bytes (IHL = 5)	Length of the IPv4 header without options

Analysis / Explanation

Because PC-A and PC-B reside on different IP networks, direct Layer-2 delivery is impossible; PC-A instead forwards the packet to its default gateway (the router). The Network layer at the router examines the destination IP address in each incoming packet, consults its routing table, and forwards the packet out of the appropriate interface toward the destination network — this process is repeated at every hop until the packet reaches PC-B. Each time the router forwards the packet, it decrements the TTL field by 1, which is why the packet reaching PC-B shows a TTL one less than the original value sent by PC-A; this mechanism prevents packets from

circulating indefinitely if a routing loop exists. The Protocol field (value 1 for ICMP in this capture) tells the receiving IP stack which upper-layer protocol to hand the payload to, while the Header Length field marks where the IPv4 header ends and the payload begins. Overall, this experiment demonstrates that the Network layer's core responsibility is logical addressing and best-path packet forwarding between different networks, independent of the underlying physical/Data-Link technology.

3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Question 3: ARP Resolution on a Multi-Host LAN

Aim

To construct a LAN with multiple hosts connected through a switch, clear the ARP cache of one host, initiate a ping to another host, and analyze the resulting ARP Request/Reply exchange captured in Wireshark.

Procedure

1. In Cisco Packet Tracer, connect three PCs (Host-1, Host-2, Host-3) to a single 2960 switch.
2. Assign IP addresses 192.168.1.1, 192.168.1.2 and 192.168.1.3 (mask /24) to Host-1, Host-2 and Host-3 respectively.
3. On Host-1, open the command prompt and clear the ARP cache using the command 'arp -d'.
4. From Host-1, ping Host-2 (ping 192.168.1.2) to trigger address resolution.
5. Capture the exchange in Wireshark and filter using 'arp' to isolate the ARP Request and ARP Reply frames.
6. Record the Sender/Target MAC and IP address fields from both the request and the reply.

Topology Diagram

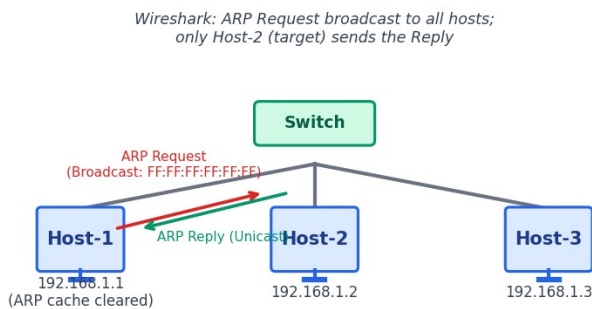


Fig. 3 — ARP Request broadcast and unicast ARP Reply on the LAN

Wireshark Observation — ARP Packet Fields

Field	ARP Request	ARP Reply
Sender MAC Address	AA:AA:AA:AA:AA:01 (Host-1)	BB:BB:BB:BB:BB:02 (Host-2)
Sender IP Address	192.168.1.1	192.168.1.2

Field	ARP Request	ARP Reply
Target MAC Address	00:00:00:00:00:00 (unknown)	AA:AA:AA:AA:AA:01 (Host-1)
Target IP Address	192.168.1.2	192.168.1.1
Destination	Broadcast (FF:FF:FF:FF:FF:FF)	Unicast (directly to Host-1)

Analysis / Explanation

With its ARP cache cleared, Host-1 has no known mapping between Host-2's IP address and its MAC address, so it cannot yet construct a valid Ethernet frame to reach it. Host-1 therefore broadcasts an ARP Request to FF:FF:FF:FF:FF:FF, asking 'Who has 192.168.1.2? Tell 192.168.1.1'. Since the frame is a broadcast, the switch floods it out of every port, so all hosts on the LAN receive and process it — but only Host-2, whose IP matches the target, replies. Host-2 responds with a unicast ARP Reply directly to Host-1's MAC address, containing its own MAC address as the answer. Host-1 then caches this IP-to-MAC mapping in its ARP table (viewable using 'arp -a') and uses it to address the subsequent ICMP Echo Request frame directly to Host-2, avoiding the need to broadcast for every future packet. This demonstrates that ARP is the essential protocol that bridges the Network layer's logical IP addressing with the Data Link layer's physical MAC addressing, enabling frame delivery within a local network.

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Question 4: ICMP Echo Request/Reply and RTT Measurement

Aim

To develop a network with two PCs connected through a router, test connectivity using ping, capture ICMP traffic in Wireshark, examine the ICMP header fields, calculate the Round Trip Time (RTT), and discuss ICMP's role in troubleshooting.

Procedure

1. Build the topology PC-A — Router — PC-B in Cisco Packet Tracer and cable the devices accordingly.
2. Configure PC-A with IP 10.0.0.1/24 and PC-B with IP 20.0.0.1/24, and configure the router's two interfaces as their respective default gateways.
3. Verify connectivity from PC-A by executing ping 20.0.0.1 and observe the reply statistics.
4. Capture the exchange in Wireshark and filter on 'icmp' to isolate Echo Request and Echo Reply packets.
5. Examine the ICMP Type, Code, Identifier and Sequence Number fields, and note the timestamps of each request/reply pair.
6. Calculate the RTT for each ping as (time reply received – time request sent).

Topology Diagram

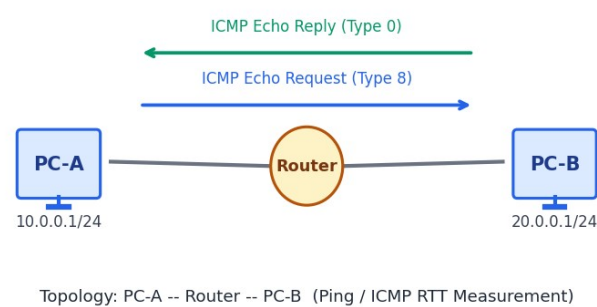


Fig. 4 — ICMP Echo Request and Echo Reply between PC-A and PC-B via the router

Wireshark Observation — ICMP Fields

Field	Echo Request	Echo Reply
Type	8	0
Code	0	0
Identifier	0x0001	0x0001 (same as request)
Sequence Number	1, 2, 3, 4 (increments per ping)	Matches the corresponding request
Sample RTT	Request sent at t = 0.000 s	Reply received at t = 0.012 s → RTT ≈ 12 ms

Analysis / Explanation

ICMP (Internet Control Message Protocol) is used by the ping utility to test reachability: PC-A sends an ICMP Echo Request (Type 8, Code 0) carrying a unique Identifier and an incrementing Sequence Number, which the router forwards toward PC-B. On receiving it, PC-B's IP stack immediately generates an ICMP Echo Reply (Type 0, Code 0) echoing back the same Identifier and Sequence Number so PC-A can match each reply to its corresponding request. The RTT is calculated by measuring the elapsed time between sending the request and receiving the matching reply; consistently low and stable RTT values indicate a healthy path, whereas high or fluctuating RTTs, timeouts, or 'Destination Unreachable'/'Time Exceeded' messages point to congestion, misconfiguration, or a broken route. Because ICMP operates directly over IP and requires no connection setup, it is a lightweight yet powerful tool for network administrators to quickly verify reachability, measure latency, and diagnose routing or connectivity failures during troubleshooting.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

Question 5: HTTP over TCP — Three-Way Handshake and Reliable Delivery

Aim

To create a client-server network in Cisco Packet Tracer, host a webpage via HTTP, access it from a client browser, capture the exchange in Wireshark, and analyze how TCP provides reliable delivery for HTTP traffic.

Procedure

- 1. Place a Server device and a Client PC in Cisco Packet Tracer and connect them via a switch/router as required.
- 2. On the server, open Services > HTTP and ensure the HTTP service is turned ON; assign it IP 192.168.10.1/24.
- 3. Configure the client PC with IP 192.168.10.2/24 and the correct default gateway.
- 4. On the client, open the Web Browser application (Desktop tab) and enter the server's IP address, e.g. http://192.168.10.1, to request the hosted webpage.
- 5. Capture the exchange in Wireshark and filter using 'tcp' and 'http' to isolate the three-way handshake, the HTTP GET request, and the HTTP response.
- 6. Examine the SYN, SYN-ACK, ACK segments followed by the HTTP GET and HTTP 200 OK response.

Topology / Message Sequence Diagram

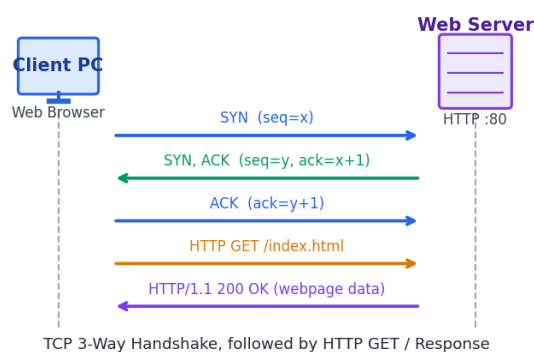


Fig. 5 — TCP three-way handshake followed by the HTTP GET request/response

Wireshark Observation — TCP/HTTP Exchange

Step	Segment / Message	Key Fields
1	SYN (Client → Server)	Seq = x, Flags = SYN
2	SYN, ACK (Server → Client)	Seq = y, Ack = x+1, Flags = SYN,ACK
3	ACK (Client → Server)	Ack = y+1, Flags = ACK — connection established
4	HTTP GET /index.html	Client requests the webpage over the established TCP session
5	HTTP/1.1 200 OK	Server responds with the requested HTML content

Analysis / Explanation

Before any HTTP data can be exchanged, TCP must establish a reliable, connection-oriented session between the client and the server using the three-way handshake: the client sends a SYN segment with an initial sequence number, the server responds with SYN-ACK acknowledging that sequence number and proposing its own, and the client completes the handshake with a final ACK. Only after this handshake does the client's browser issue the HTTP GET request over the now-established connection, and the server replies with an HTTP 200 OK response carrying the webpage content. TCP guarantees reliability throughout this exchange in three ways: every segment carries a sequence number so the receiver can reorder segments that arrive out of order and detect any that are missing; the receiver sends acknowledgment (ACK) numbers back to confirm safe receipt, causing the sender to retransmit any segment that is not acknowledged within a timeout; and flow-control windowing prevents a fast sender from overwhelming a slow receiver. Because HTTP relies entirely on TCP for these guarantees, the application layer itself does not need to worry about lost, duplicated, or misordered data — it simply hands data to TCP and trusts that it will arrive intact and in order at the other end.

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Q4					
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